Practice workbook · Explain the liver's roles in deamination and breaking down drugs

Deamination of surplus amino acids to urea, why kidney patients limit protein, the liver's large blood supply, and why liver damage lets alcohol or medicines build up.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-LIVER-DEAMINATION-AND-DETOXIFICATION

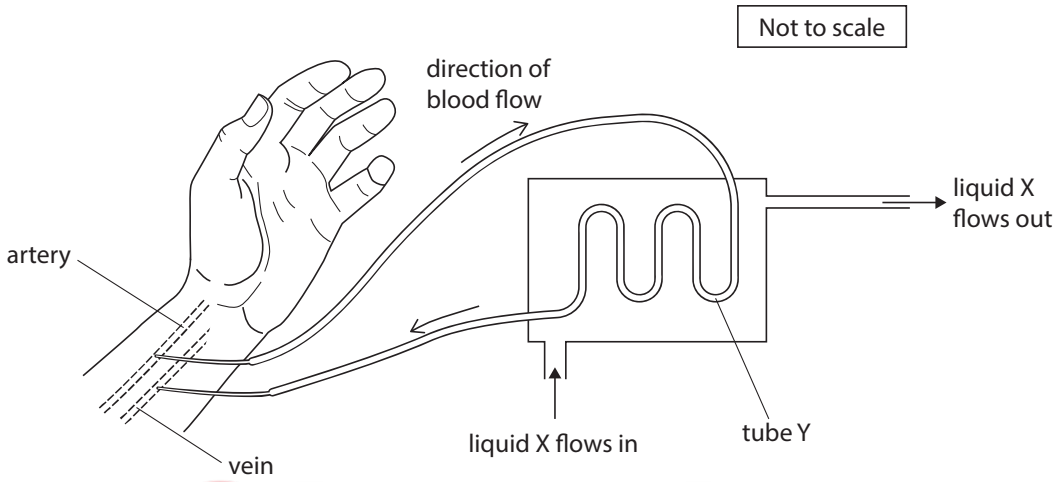
1. Question 5 · part c · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part c only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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1. Question 5 · part c · 4 marks

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Worked-example retry · Attempt part c only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

(2)

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

(1)

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

(3)

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1. Question 5 · part c · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part c only.

(c) Explain why people who have to use kidney dialysis machines are often told to eat less protein in their diet.

(4)

(Total for Question 5 = 15 marks)

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Working and answer

2. Question 8 · part a.i · 3 marks

June 2019 Paper 2R

Further practice · Attempt part a.i only.

8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

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2. Question 8 · part a.i · 3 marks · Working and answer

Working and answer

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3. Question 9 · part a.i · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

(3)

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3. Question 9 · part a.i · 3 marks · Working and answer

Working and answer

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4. Question 7 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

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7 The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

(1)

.....

(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

(3)

	Structure		
Function	**glomerulus**	**proximal tubule**	**renal vein**
regulation of water level			
ultrafiltration			
reabsorption of glucose			
removal of blood from kidney			
supply of blood to kidney			
removal of urine from collecting duct			

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4. Question 7 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(b) The table shows the percentage composition of two urine samples taken from one person.

Not all of the components are shown.

The first sample was taken at midday and the second sample was taken four hours later.

Component	Percentage composition	
	first sample	second sample
ammonia	0.05	0.08
uric acid	0.03	0.08
ions	1.34	1.46
urea	2.00	2.80
water	95.00	91.50

(i) Describe how urea is produced in the body.

(2)

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4. Question 7 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) The ADH response changes the percentage composition of urine.

Explain the ADH response that occurred in the person's body between the first and the second sample being collected.

(5)

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(Total for Question 7 = 11 marks)

4. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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5. Question 7 · part b · 3 marks

June 2023 Paper 1

Further practice · Attempt part b only.

7 A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 cm^3 per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GFR in cm^3 per minute	80	75	79	79	67	60	50	45	35

(a) (i) Plot a line graph of the data on the grid.

(4)



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5. Question 7 · part b · 3 marks

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Further practice · Attempt part b only.

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(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

(2)

(iii) A person needs dialysis if the GFR falls to 15 cm^3 per minute.

Use the data to estimate when the person will probably need to start dialysis.

(2)

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(iv) Discuss the accuracy of this estimate.

(3)

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5. Question 7 · part b · 3 marks

June 2023 Paper 1

Further practice · Attempt part b only.

(b) During the early stages when a person's kidneys are not working correctly, they are advised to control the amount of protein in their diet.

Explain why the protein in their diet should be controlled.

(3)

(Total for Question 7 = 14 marks)

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5. Question 7 · part b · 3 marks · Working and answer

Working and answer

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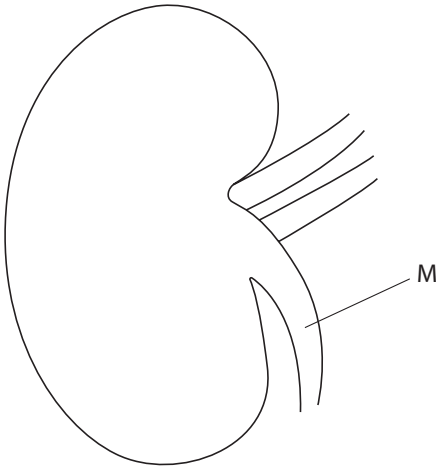
6. Question 7 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

6. Question 7 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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6. Question 7 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

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(ii) Which is the main nitrogen containing excretory product produced from the breakdown of protein in the diet?

Use the information in the table to explain your answer.

(2)

(iii) The breakdown of muscle tissue continues at a constant rate and is not affected by diet.

Discuss whether the information in the table supports this statement.

(4)

(Total for Question 7 = 14 marks)

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6. Question 7 · part c.ii · 2 marks · Working and answer

Working and answer

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7. Question 8 · part b · 4 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

8 The table shows the rate of blood flow through different organs when a person is at rest. The table also shows the masses of these organs.

Organ	Mass in kg	Rate of blood flow in cm^3 per minute	
		through the whole organ	per 100g of the organ
kidneys	0.3	1200	
skeletal muscles	25.0		3
liver	1.5	1500	100

(a) Calculate the missing values and put them in the table.

(4)

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(b) Explain why the liver has a high rate of blood flow.

(4)

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7. Question 8 · part b · 4 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

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(c) Explain any changes that would occur in the blood flow through these organs during vigorous exercise.

(4)

(Total for Question 8 = 12 marks)

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Working and answer

8. Question 4 · part b.i · 4 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

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8. Question 4 · part b.i · 4 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

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8. Question 4 · part b.i · 4 marks · Working and answer

Working and answer

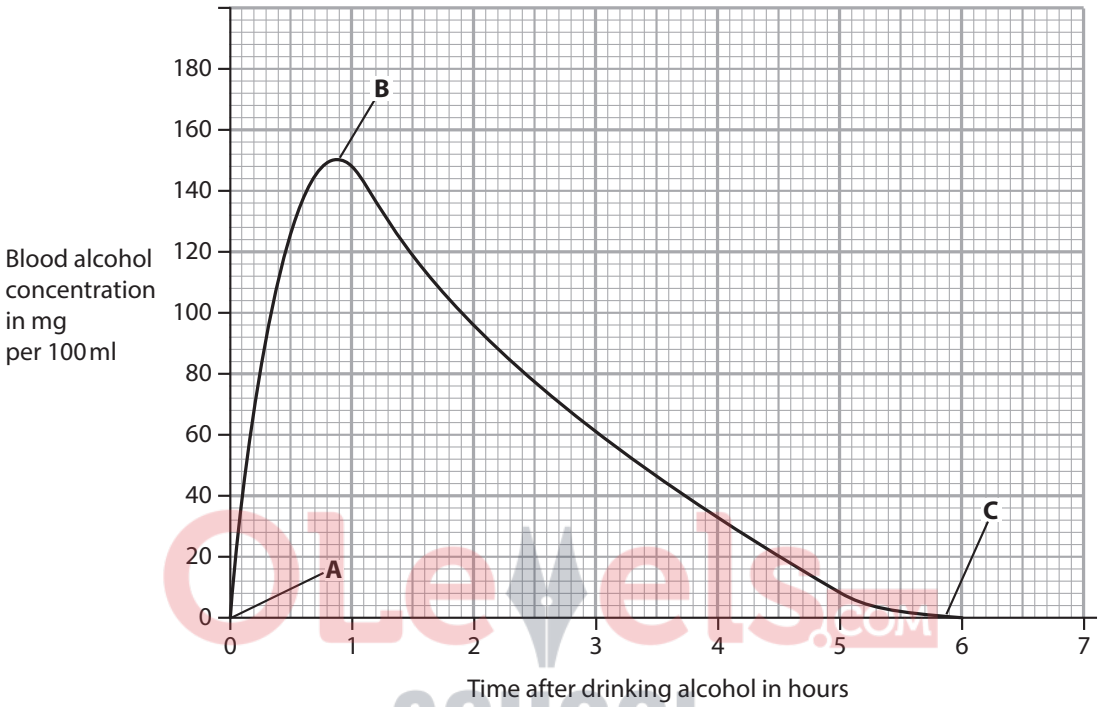
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9. Question 6 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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9. Question 6 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

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(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

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9. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

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10. Question 5 · part e · 2 marks

June 2025 Paper 2

Further practice · Attempt part e only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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10. Question 5 · part e · 2 marks

June 2025 Paper 2

Further practice · Attempt part e only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

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(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

10. Question 5 · part e · 2 marks · Working and answer

Working and answer

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